

EXP 8 - FABRIC NETWORK DEPLOYMENT AND MANAGEMENT

AIM

To deploy, manage, monitor, and maintain a Hyperledger Fabric network by creating the network, verifying network components, deploying chaincode, monitoring Docker containers, and performing network shutdown and cleanup operations.

SOFTWARE REQUIREMENTS

- Kali Linux / Ubuntu
- Docker & Docker Compose
- Node.js (Version 18 or later) & npm
- Git
- Hyperledger Fabric Samples, Binaries, and Docker Images
- Visual Studio Code (Optional)

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

THEORY

Hyperledger Fabric network deployment involves setting up the blockchain infrastructure required for enterprise applications. A Fabric network consists of peer nodes, ordering service nodes, Certificate Authorities (CAs), channels, chaincode, and client applications.

Network management encompasses:

- Starting and stopping network nodes using Docker container orchestration
- Creating and joining application channels
- Deploying and packaging smart contracts (chaincode)
- Monitoring container state, health metrics, and component logs
- Inspecting peer, CA, and Raft orderer services
- Reclaiming Docker system storage and cleaning temporary cryptographic artifacts

Proper deployment and lifecycle management guarantee transaction availability, data durability, and clear operational visibility.

PROCEDURE

- **Step 1: Navigate to the Fabric Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

- **Step 2: Start the Fabric Network with CAs and Application Channel**

`./network.sh up createChannel -ca`

- **Step 3: Verify Running Docker Containers**

`docker ps`

- **Step 4: Display Pulled Fabric Docker Images**

`docker images`

- **Step 5: Check Network Status Across All Containers**

`docker ps -a`

- **Step 6: Deploy the Asset Transfer Chaincode**

`./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript`

- **Step 7: Verify Installed Chaincode Package on Peer**

`peer lifecycle chaincode queryinstalled`

- **Step 8: Monitor Peer Node Logs**

`docker logs --tail 10 peer0.org1.example.com`

- **Step 9: Monitor Orderer Service Logs**

`docker logs --tail 10 orderer.example.com`

- **Step 10: Verify Joined Channels on Peer**

`peer channel list`

- **Step 11: Stop the Fabric Network**

`./network.sh down`

- **Step 12: Clean Unused Docker System Resources**

`docker system prune -f`

OUTPUT / SCREENSHOTS

Screenshot 1: Starting Network Infrastructure & Channel Creation

```
kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel
Generating certificates using Fabric CA
Generating certs for Org1...
Generating certs for Org2...
Generating certs for Orderer Org...
Creating channel 'mychannel'... done
Starting peer0.org1.example.com... done
Starting peer0.org2.example.com... done
Starting orderer.example.com... done
Setting anchor peer for org1...
Setting anchor peer for org2...
Channel 'mychannel' joined
Fabric Network Started Successfully
```

Screenshot 2: Container Status and Image Verification

```
kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ docker ps --format "table
CONTAINER ID    IMAGE                                COMMAND                                STATUS
8a2e1f4091c0    hyperledger/fabric-peer:latest      "peer node start"                    Up 2 minutes
3c9d4b8102a1    hyperledger/fabric-peer:latest      "peer node start"                    Up 2 minutes
d4e21a8f90c3    hyperledger/fabric-orderer:latest   "orderer"                            Up 2 minutes
21f4c78a90e4    hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se..."           Up 2 minutes
45e89a01f8d2    hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se..."           Up 2 minutes
b8912c0192e1    hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se..."           Up 2 minutes

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ docker images --format "table
REPOSITORY          TAG    SIZE
hyperledger/fabric-peer    latest    162MB
hyperledger/fabric-orderer  latest    108MB
hyperledger/fabric-ca      latest    238MB
hyperledger/fabric-tools   latest    502MB
```

Screenshot 3: Network Teardown and Cleanup

```
Stopping peer0.org2.example.com... done
Stopping orderer.example.com... done
Stopping ca_org1... done
Stopping ca_org2... done
Stopping ca_orderer... done
Removing containers... done
Removing generated artifacts and crypto material... done
Removing network net_test... done
Network stopped successfully

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ docker system prune -f
Deleted Containers:
8a2e1f4091c0e3a479
3c9d4b8102a1c0d481
d4e21a8f90c379a25b

Deleted Networks:
net_test

Total reclaimed space: 142.6MB
```

RESULT

The Hyperledger Fabric network was successfully deployed and managed. The blockchain network components, including peer nodes, ordering service, Certificate Authorities, and application channel, were verified. Chaincode was deployed successfully, network status was monitored, and the Fabric network was shut down and cleaned up successfully.